

Assignment - 03

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Waste to Energy Facility for Mumbai (1000 TPD) using incineration, gasification and pyrolysis.

Composition - Municipal waste composition directly affects conversion efficiency and plant design. The feedstock data to plant is derived from surveys of CPCB and BMC.

Component	% by Wt	Moisture Content (%)	LHV (MJ/kg)
Biodegradable	45	55	6
Plastics and Rubber	12	5	25
Paper and Cardboard	10	10	14
Textiles and Leather	8	10	15
Wood and Yard Waste	5	30	12
Metals	5	-	-
Glass and Ceramics	5	-	-
Inert / Fine Dust	10	20	1.5
	100	45	

Weighted avg. lower heating value (LHV) is approx.

7.5 MJ/kg for thermochemical conversion after minor drying and segregation.

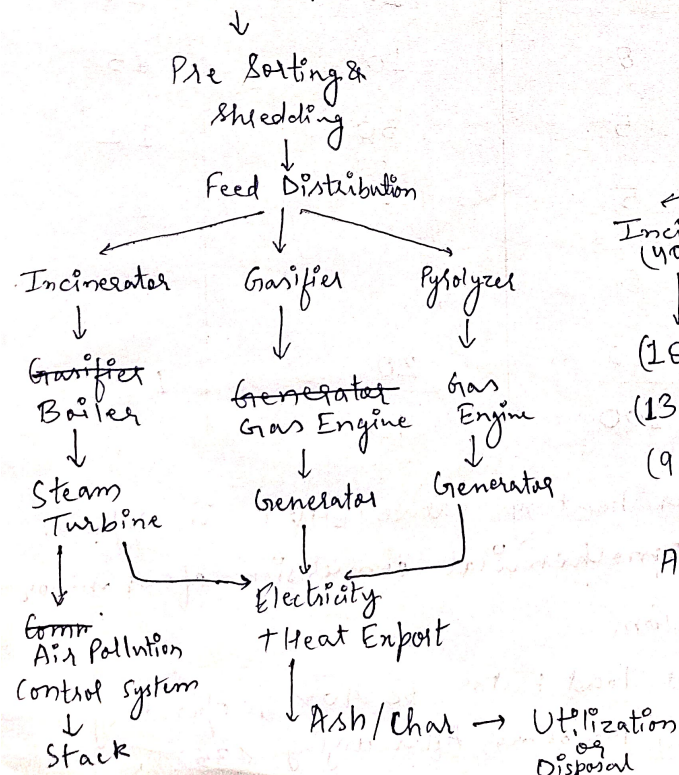
Assuming that Plant load factor be 90%, operating hours be 20 h/day, Moisture After drying < 35%.

Feed Allocation to incineration (40%), gasification (35%),
pyrolysis (25%), steam generation at
40 bar, 400°C, Electricity tariff ₹6/kWh with a discount
rate of 10%. having plant life 20 years.

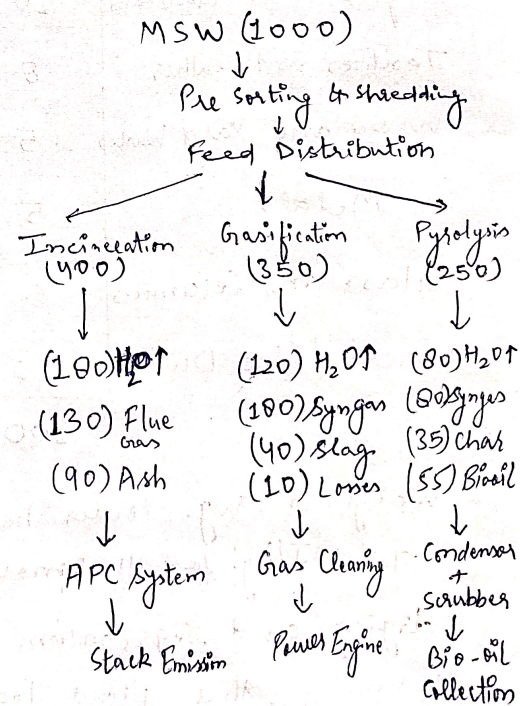
The facility is designed as an integrated system
with common waste preprocessing, utilities and emission
control. The MSW is segregated, shredded and converted
into Refuse ~~derived fuel~~ suitable for thermochemical
preprocessing as -

- Incineration line (400 TPD) - complete combustion for
steam and power.
- Gasification line (350 TPD) - partial oxidation producing
syngas.
- Pyrolysis line (250 TPD) - thermal cracking producing
gas, oil, and char.

Block Flow Diagram MSW (Input)



Mass Flow Diagram



Incineration line - Waste is fed into grate type incinerator operating at $850-1050^{\circ}\text{C}$ under excess air ($\lambda=1.6-2.0$).

Combustion reaction release thermal energy, producing high temperature flue gas used to generate steam at 40 bar, 400°C . Steam drives a turbine generator system, producing approximately 22MW net electricity.

Gasification line - In fixed-bed gasifier, MSW undergoes partial oxidation at $800-950^{\circ}\text{C}$ with limited air ($\lambda=0.3$). The resulting syngas ($\text{CO}, \text{H}_2, \text{CH}_4, \text{CO}_2, \text{N}_2$) is cooled and cleaned using venturi, scrubber, bag filter and condenser and then used in gas engines or turbines to produce $\sim 21\text{MW}$ of power. Slag produced is environmentally inert.

Pyrolysis line - MSW with high plastic and rubber content undergoes thermal decomposition in absence of oxygen in a rotary kiln at $450-700^{\circ}\text{C}$. The pyrolysis vapors are condensed into a bio-oil (35MJ/kg) while remaining syngas is combusted in a gas engine to generate $\sim 18\text{MW}$ of power. Char (20-25%) is collected for use in cement or agriculture.

values assumed here for energy generation are based on reports of MNRE, TERI. These studies indicate typical electrical yields ($550-900\text{ kWh/tonne}$)

Energy Generation and Efficiency Summary

Technology	Feed (TPD)	Net Power (MW)	Electricity Generation Efficiency
Incineration	400	22	20-25%
Gasification	350	21	25-30%
Pyrolysis	250	18	15-20%
Total	1000	67	

Energy Production and Cost Estimation

Technology	Energy Output (kWh/tonne)	Estimated Unit Cost of Electricity (£/kWh)	Estimated unit cost of Heat (£/MJ)
Incineration	~600-700	14-17	0.5-0.7
Gasification	~800-900	12-15	0.4-0.6
Pyrolysis	~500-600	16-19	0.6-0.8

Capital and Operational Expenditures -

	Capital	O&M	Land Requirement
Incineration	£5 crore/MW = $5 \times 22 = 110$ crore	£25-30 crore/year	6-8 acres
Gasification	£4.5 crore = $4.5 \times 27 = 121.5$ crore	£20-25 crore/year	5-7 acres
Pyrolysis	£3.5 crore = $3.5 \times 18 = 63$ crore	£20 crore/year	4-6 acres

Annual Electricity Output = Power $\times$ Work hrs

Unit Cost of Electricity = $\frac{\text{Cost capital} + \text{Cost O\&M}}{\text{Annual}}$

Capital Recovery Factor = $\frac{0.1 \times (1.1)^{10}}{(1.1)^{10} - 1} = 0.11746$

Technology	Cost of Electricity (£/kWh)	Heat cost (energy proportional) (£/MJ)
Incineration	7.8	0.49
Gasification	5.9	0.37
Pyrolysis	7.0	0.59

References -

- CPCB (2023), Waste to Energy Projects in India.
- NITI Aayog & MNRE (2022), Waste to Energy: A Way Forward
- TERI (2021), Feasibility Analysis of Waste to Energy Technologies in India.
- IEA Bioenergy (2021), ~~Bio~~ Waste to Energy: State of the Art Report.
- CEA (2023), Renewable Integration Report